

GenAI Powered Data Analytics Job Simulation

Project Summary | Tasks 1–4

500

Customers

19

Variables

80

Delinquent

16.0%

Delinquency

Project Overview

This summary consolidates Tasks 1–4: exploratory analysis and risk profiling, a conceptual delinquency-prediction framework, stakeholder-ready business storytelling, and an AI-driven collections strategy.

Four-Task Journey

Task 1 — Exploratory Data Analysis & Risk Profiling: Understand customer risk, data quality and delinquency patterns.

Task 2 — Predicting Delinquency with AI: Design an interpretable framework for delinquency probability and prioritization.

Task 3 — Business Report & Data Storytelling: Translate findings into a SMART collections recommendation and responsible-AI plan.

Task 4 — AI-Driven Collections Strategy: Implement a governed SENSE → DECIDE → ACT → LEARN operating model.

Core Business Objective

Use data and AI to prioritize appropriate collections interventions earlier while protecting fairness, transparency, privacy and human oversight.

01 | Exploratory Data Analysis & Risk Profiling

Task 1 established the analytical foundation for delinquency prediction by assessing structure, completeness, consistency, exploratory risk signals and modeling considerations.

Objective	Assess dataset structure, completeness, consistency and exploratory risk signals before predictive modeling.
Dataset	500 customers across 19 variables; 9 numerical and 10 categorical/text variables; no duplicate rows.
Missing data	Income 39 (7.8%); Loan Balance 29 (5.8%); Credit Score 2 (0.4%); total missing cells 70/9,500 (0.74%).
Risk signals	Credit Utilization: 50.7% delinquent vs 48.9% non-delinquent. DTI: 30.6% vs 29.7%.
Anomalies	4 records (0.8%) had Credit Utilization above 100%; Employment Status labels need standardization.
Top 3 candidates	Credit Utilization, Debt-to-Income Ratio and Credit Score; predictive strength should be validated through modeling.

Illustrative Segment Risk Signals

These rates are descriptive, not causal, and should be validated on larger representative data before being used for collections decisions.

Segment	Delinquency Rate
Business card	21.3%
Los Angeles	19.6%
Unemployed	19.4%

Key Takeaway

Credit Utilization, Debt-to-Income Ratio and Credit Score are the three leading candidate risk indicators. Simple correlations were weak, so predictor importance should be established through out-of-sample modeling.

02 | Predicting Delinquency with AI

Task 2 translated the EDA findings into a conceptual predictive framework. The objective is prioritization—not automatic adverse action.

Objective	Predict probability of Delinquent_Account = 1 so Collections can prioritize appropriate early interventions.
Primary model	Logistic Regression as an interpretable baseline with probability output and reviewable coefficients.
Challenger	Decision Tree to capture nonlinear relationships and interactions, with tuning to reduce overfitting.
Top 5 inputs	Credit Utilization, Debt-to-Income Ratio, Credit Score, Income and Loan Balance.
Workflow	Preprocess → training-set-only imputation → validate anomalies → stratified train/validation/test → model → probability → threshold → risk priority → human-reviewed intervention.
Evaluation	ROC-AUC, precision, recall, F1, confusion matrix, calibration, fairness, stability and out-of-sample performance.

Model Logic

Customer inputs → preprocessing → Logistic Regression probability → validated operating threshold → risk priority → human-reviewed intervention.

Evaluation & Governance

- Use stratified train/validation/test splits because delinquency represents 16.0% of the supplied dataset.
- Evaluate ROC-AUC, precision, recall, F1, confusion matrix and probability calibration.
- Compare relevant group performance and calibration where legally and operationally appropriate.
- Select the final model using out-of-sample performance, calibration, stability and fairness—not accuracy alone.
- Protect confidentiality and avoid unnecessary personal or sensitive financial data in GenAI workflows.

Important Limitation

Task 2 provides a modeling plan and evaluation strategy; it does not report final predictive performance or claim a production-ready model.

03 | Business Report & Data Storytelling

Insight	Elevated Credit Utilization is a higher-priority signal for delinquency-risk assessment, alongside DTI and Credit Score.
Recommendation	Run a 6-week targeted collections outreach pilot for customers in the validated high-risk model band with elevated Credit Utilization, with human review.
Specific	Use supportive early contact; do not use the score as an automatic adverse action.
Measurable	Target a 10% relative reduction in 30-day delinquency versus baseline/comparable control; track contact, cure and escalation rates.
Actionable	Use model probability and an agreed operating threshold to prioritize accounts and document outcomes.
Time-bound	Operate for 6 weeks, review weekly, and complete a formal outcome assessment at the end.

SMART Recommendation

Run a 6-week targeted outreach pilot for the validated high-risk model band with elevated Credit Utilization and human review. Target a 10% relative reduction in 30-day delinquency versus baseline/comparable control.

04 | Implementing an AI-Driven Collections Strategy

Operating loop	SENSE customer/repayment signals → DECIDE risk score + business rules → ACT approved outreach → LEARN from outcomes and monitoring.
Agentic AI	Automate routine data ingestion, scoring, reminders/follow-ups, routing, logging and exception surfacing within pre-approved policies.
Human control	People retain control over hardship/personalized plans, borderline or high-impact cases, complaints/disputes, fairness/drift reviews and material model/policy changes.
Guardrails	Fairness, explainability, compliance/privacy, human oversight, audit logs, and continuous monitoring of performance, calibration, fairness, drift, complaints and overrides.
Impact	Pilot target: 10% relative reduction in 30-day delinquency over 6 weeks; also track cure, productivity, cost per cured account and customer outcomes.
Deployment	Validate → Pilot → Measure → Govern. Scale only when business, fairness, compliance and governance gates are met.

Responsible AI Principles

- Fairness: monitor model error rates and calibration across relevant customer groups.
- Explainability: provide interpretable outputs, plain-language reasons, documented features, thresholds and limitations.
- Compliance & Privacy: apply applicable requirements and minimize unnecessary data.
- Human Oversight: review consequential actions, hardship cases, exceptions, complaints and disputes; maintain auditability.

Executive Takeaway

Build toward governed autonomy: let AI prioritize and coordinate routine collections, while people retain control over consequential customer decisions. Deployment path: VALIDATE → PILOT → MEASURE → GOVERN.

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